



MONGODB 102 Workshop

Exercise 1

Exercise 1 - MongoDB Aggregation with JSON

Overview

This exercise is to use the aggregation framework to perform queries that require some computation (e.g. sum, count, etc).

[Aggregation](#) in MongoDB allows for the transforming of data and results in a more powerful fashion than from using the find() command.

Through the use of multiple stages and expressions, you are able to build a "pipeline" of operations on your data to perform queries that require aggregation functions like sum, average, max, min, etc and analytic operations.

The stages in a pipeline can filter, sort, group, reshape and modify documents that pass through the pipeline.

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db.collection.aggregate()

This command calculates aggregate values for the data in a collection or a view.

1. If your mongosh is closed, Open your terminal or command prompt to run the connection string or open [embedded mongosh](#) in MongoDB Compass.
2. Please use the **connection string** in **Environment Setup**. It will prompt you for the password created in Environment Setup.
3. Type “**show dbs**” to get the list of databases that were loaded earlier. If you are using an EA cluster in your organization, it will be different output. Just make sure “**sample_airbnb**” and “**sample_supplies**” are in the list.

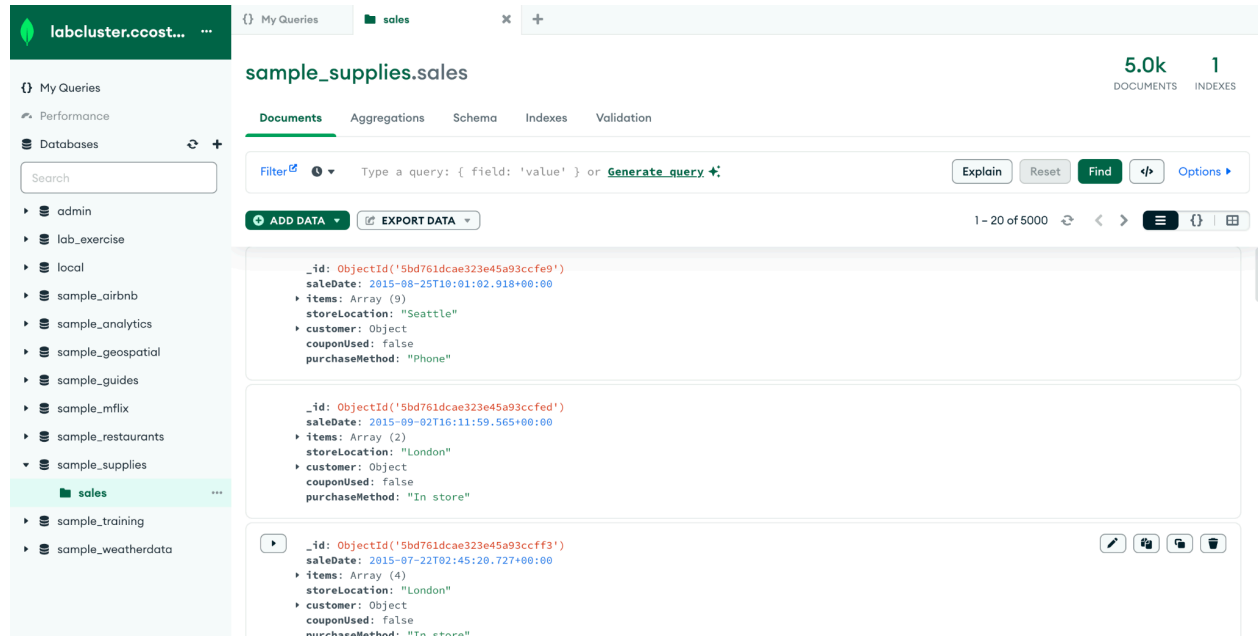
```
Last login: Sun Jan 28 14:39:22 on ttys001
limlikhia@Lims-MacBook-Pro ~ % mongosh "mongodb+srv://labcluster.ccostox.mongodb.net/" --apiVersion 1 --username jas24061
Enter password: *****
(Current Mongosh Log ID: 65b5f7eef09397a98ee5eebb
Connecting to:   mongodb+srv://<credentials>@labcluster.ccostox.mongodb.net/?appName=mongosh+2.1.1
Using MongoDB:  6.0.13 (API Version 1)
Using Mongosh:  2.1.1

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

Atlas atlas-2goyid-shard-0 [primary] test> show dbs
lab_exercise      8.00 KiB
sample_airbnb     52.76 MiB
sample_analytics  9.56 MiB
sample_geospatial 1.24 MiB
sample_guides     40.00 KiB
sample_mflix      114.90 MiB
sample_restaurants 6.33 MiB
sample_supplies   1.05 MiB
sample_training   49.19 MiB
sample_weatherdata 2.60 MiB
admin             280.00 KiB
local             2.73 GiB
Atlas atlas-2goyid-shard-0 [primary] test> █
```

4. In this exercise, you will use multiple databases and collections to do the aggregation.
5. Type “**use sample_supplies**” in the terminal. Before that, let’s see the data in Compass. It is a list of documents that contains information about the sales in different store locations and each having a different purchase method. We will use the aggregation framework to find the data in subsequent steps.

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6. The command below is to count the number of sales in the **‘Denver’** group by the **‘purchaseMethod’**. The result will be sorted by the number of sales in descending order.

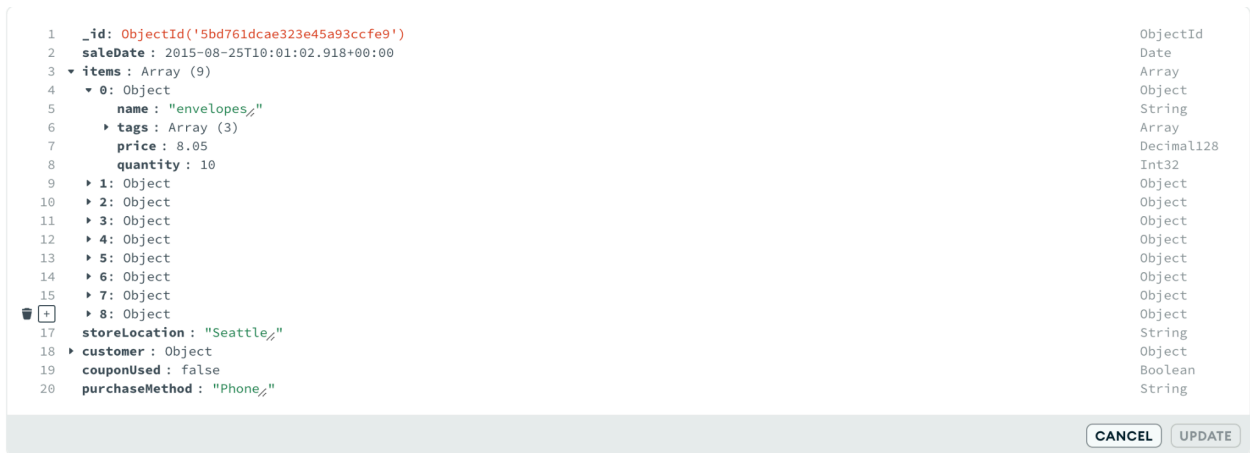
```
db.sales.aggregate([
  { $match: { 'storeLocation': 'Denver' } },
  { $group: { '_id': "$purchaseMethod", total: { $sum: 1 } } },
  { $sort: { total: -1 } },
  { $project: { '_id': 0, purchaseMethod: "$_id", total: 1 } }
])
```

Output:

```
[
  { total: 864, purchaseMethod: 'In store' },
  { total: 513, purchaseMethod: 'Online' },
  { total: 172, purchaseMethod: 'Phone' }
]
```

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7. Next, you are going to calculate the total of each sales by sum up the total price [price * quantity] of each item in the “items” array.



```
1  _id: ObjectId('5bd761dcae323e45a93ccfe9')
2  saleDate: 2015-08-25T10:01:02.918+00:00
3  items: Array (9)
4    0: Object
5      name: "envelopes"
6      tags: Array (3)
7      price: 8.05
8      quantity: 10
9    1: Object
10   2: Object
11   3: Object
12   4: Object
13   5: Object
14   6: Object
15   7: Object
16   8: Object
17  storeLocation: "Seattle"
18  customer: Object
19  couponUsed: false
20  purchaseMethod: "Phone"
```

ObjectId
Date
Array
Object
String
Array
Decimal128
Int32
Object
Object
Object
Object
Object
Object
Object
String
Object
Boolean
String

CANCEL UPDATE

8. To make things simple to understand, you will do this with one record. The command below is to find the document based on its “_id” and count the total price of sales by purchaseMethod. You can sum up the total manually to check if it is correct. [\\$unwind](#) deconstructs an array field from the input documents to output a document for each element.

```
db.sales.aggregate([
  { $match: { 'storeLocation': 'Seattle', '_id': ObjectId('5bd761dcae323e45a93ccfe9') } },
  { $unwind: "$items" },
  { $group: { '_id': "$purchaseMethod", total: { $sum: { $multiply: ["$items.price", "$items.quantity"] } } } }
])
```

Output:

```
[ { _id: 'Phone', total: Decimal128('4405.57') } ]
```

9. The command below is to count the total price of sales in ‘Seattle’ group by purchaseMethod

```
db.sales.aggregate( [
  { $match: { 'storeLocation': 'Seattle' } },
  { $unwind: "$items" },
  { $group : { _id : "$purchaseMethod", total : { $sum : { $multiply : ["$items.price", "$items.quantity"] } } },
  { $sort : {total:1} }
])
```

Output:

```
[
  { _id: 'Phone', total: Decimal128('280599.75') },
  { _id: 'Online', total: Decimal128('658850.42') },
]
```

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```
{ _id: 'In store', total: Decimal128('1316497.52') }
]
```

10. Type “**use sample_airbnb**” in the terminal.

11. The command below is to filter and extract certain fields from ‘**listingsAndReviews**’ collections and output them as separate collections named as ‘**listingsAndReviews2**’.

```
db.listingsAndReviews.aggregate([
  { $match : { 'accommodates' : { '$gte' : 8 } } },
  { $project : { '_id': 0, 'listing_url': 1, 'name': 1, 'summary': 1, 'property_type': 1, 'accommodates': 1 } },
  { $out: { 'db': 'sample_airbnb', 'coll': 'listingsAndReview2' } }
])
```

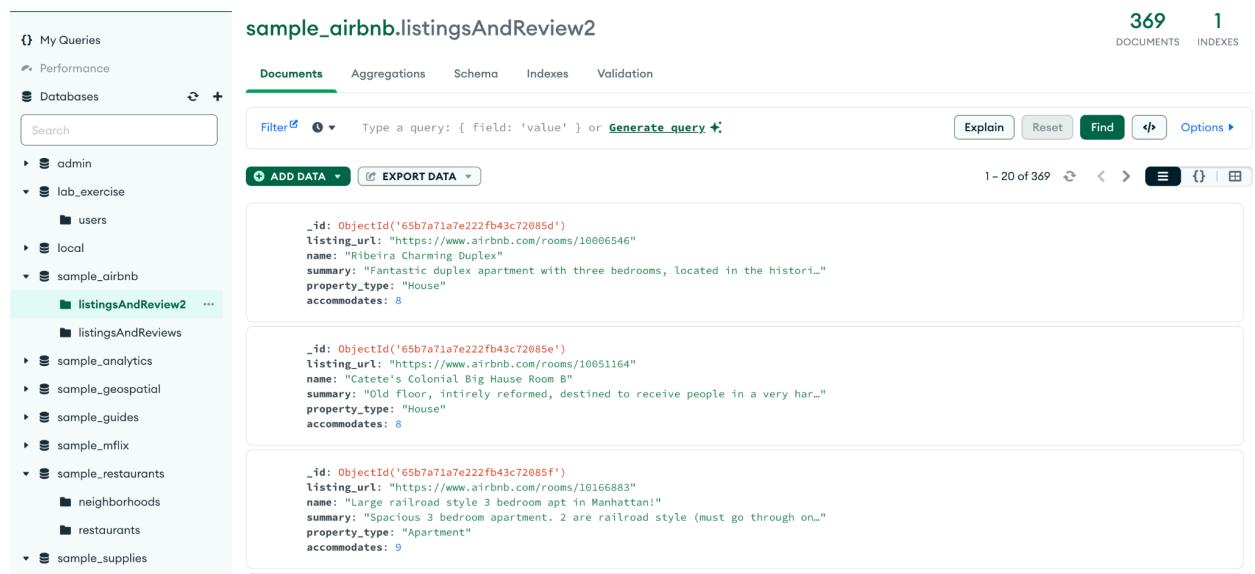
12. The command below is to verify ‘**listingsAndReview2**’ is created.

```
show collections
```

Output:

```
listingsAndReview2
listingsAndReviews
```

13. Go back to the **compass** and refresh . Select the collection as shown below. You will see the new collection.



The screenshot shows the MongoDB Compass interface. On the left sidebar, the 'sample_airbnb' database is selected, and the 'listingsAndReview2' collection is highlighted. The main panel displays the 'sample_airbnb.listingsAndReview2' collection with 369 documents and 1 index. The 'Documents' tab is active, showing a list of document snippets. Each snippet includes fields like '_id', 'listing_url', 'name', 'summary', 'property_type', and 'accommodates'. For example, the first document has a listing URL 'https://www.airbnb.com/rooms/10006546' and is a 'House' with 8 accommodations.